

Research Statement

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I am a behavioral economist studying how boundedly rational behavior shapes competition and coordination. I combine game theory and laboratory experiments to examine how people acquire and process information and reason about others under different information structures and strategic environments. My goal is to use this understanding to inform the design of market mechanisms.

1. Consumer Search and the Market for Prominence

My job market paper, “Search Frictions and Welfare in a Market for Prominence: Theory and Experiment,” studies how consumer search costs affect the level and distribution of surplus when sellers compete for prominence. In online marketplaces, consumers can often obtain price information with little effort, but learning whether a product suits their tastes or needs remains costly. Consumers may stop before inspecting every product, giving sellers a reason to pay for search ads and sponsored listings that make their products easier to discover and inspect.

I develop a model in which two *ex-ante* identical sellers bid for a prominent position in a second-price auction and then simultaneously post prices. The buyer observes both prices and learns her match value for the prominent seller’s product for free. She then decides whether to incur a search cost to inspect the other seller’s product. Because prices are visible, sellers can use them strategically to influence whether the buyer searches further. I test the model’s predictions for welfare and behavior in a laboratory experiment by varying search cost between low and high levels.

In the laboratory, I find systematic departures from the predicted distribution of surplus, which I explain using observed bidding, pricing, and search behavior. Platform auction revenue changes little as search cost rises, despite the predicted increase, reflecting the weak response of the losing bids that determine auction payments. Consumer surplus exceeds equilibrium predictions at both costs and falls proportionally less with search cost than predicted. Sellers, however, jointly retain around 20 percent of total surplus after paying for prominence. One reason is that prominent sellers charge lower prices than predicted, benefiting buyers but reducing revenue per sale. These lower prices also encourage buyers with a good initial match to buy from prominent sellers without further search, helping them save the cost of another inspection. Moreover, high prices charged by non-prominent sellers discourage further search by buyers who find the prominent product a bad match, leading some to exit without purchasing from either seller. Together, these patterns explain why combined sales revenue falls below equilibrium, while payments to the platform further reduce profits.

I plan to extend this work in two directions, both theoretically and experimentally. First, I will allow the platform to choose search cost and examine how its incentives depend on both prominence payments and other revenue sources, such as sales commissions. I will study when the platform benefits from making search easier or harder, and when its preferred search cost differs from the socially optimal level. Second, I will compare alternative mechanisms for allocating prominence, including different auction formats, to identify which achieve the highest efficiency when position values depend on subsequent pricing and consumer search.

2. Depth of Strategic Reasoning

My work on bounded rationality examines where people start when reasoning about others and how far they proceed. Understanding these limits helps explain players' initial responses to unfamiliar or complex decision environments. In joint work, "Revisiting the Intra-Team Communication Method to Elicit Level-k Reasoning in Beauty Contests and 11–20 Games," we provide the first replication of Burchardi and Penczynski's beauty-contest experiment and extend their intra-team communication method to variants of the 11–20 game. Participants explain their choices in incentivized messages to teammates, allowing us to infer what they believe a non-strategic player would do and how many reasoning steps they take from that starting point.

In the beauty contest, elicited beliefs about non-strategic play are consistent with the observed behavior of players classified as non-strategic, as in the original study. However, this belief-action consistency breaks down in the reordered 11–20 game: non-strategic choices are more dispersed than higher-level players expect, and both choices and beliefs depart from the standard assumption about non-strategic play. Moreover, using our within-subject design, we find that individuals' reasoning levels inferred from messages show positive correlations across games, a relationship not revealed by choices alone. This provides tentative support for viewing depth of reasoning as a stable individual characteristic across strategic settings.

3. Communication and Coordination

In a third project, I study how the rules of communication affect coordination when players face asymmetric information about the state and uncertain signal delivery. In "Coordination under Uncertainty and Noisy Communication: A Generalized Email Game," one player knows whether the state favors a risky joint action, while the other player does not. In the favorable state, a single signal can reveal the state to the uninformed player, but has a fixed probability of failing to arrive. The model permits both an inefficient equilibrium in which the signal is ignored and a more efficient one in which players act on it, making equilibrium selection the central empirical question.

In the laboratory, I vary the communication protocol to examine how voluntary sending and its cost affect coordination. I find that relative to a protocol that sends the signal automatically, free voluntary sending by the informed player makes players more likely to take the risky action after favorable information. Adding a sending cost, however, deters informed players from sending and reduces overall action on favorable information, despite a slight increase in recipients' responsiveness to delivered signals. These results suggest that choosing to send can convey an intention to coordinate when players' interests are aligned, even when uncertain signal delivery prevents the state from becoming common knowledge. A fee intended to enhance signal credibility can nevertheless backfire by discouraging communication in the first place.

4. Research Pipeline

My future research will examine when behavioral departures from equilibrium change which market mechanisms perform best. I will compare alternative allocation and information rules in theory and in the laboratory, assessing both efficiency and the distribution of surplus. Using observed choices and elicited beliefs, I will investigate why mechanisms perform differently and test changes to their design. In platform markets, these comparisons can inform policies governing paid prominence and the product information available to consumers.